DEPARTMENT OF COMPUTING CS-212:

ASSIGNMENT 02

Object Oriented Programming Class:

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END SEMESTER PROJECT PROPOSAL :

COLOR CONTRAST AND ACESSIBILITY CHECKER

Problem statement: Visual accessibility is a problem faced globally. The World Health Organization states that there are currently 2.2 billion people worldwide suffering from some form of visual impairment. The main problems faced while viewing digital content are color blindness, low contrast sensitivity for aging populations, and difficulty in differentiating foreground text and background colors for poorly designed interfaces.

The World Wide Web Consortium published guidelines for web content accessibility, known as the Web Content Accessibility Guidelines (WCAG) version 2.1, which have set contrast ratio thresholds for all digital interfaces to be followed for accessibility. The present contrast checker tool are web based requiring stable internet connection ,are large and platform dependent and not available for educational purposes and modifications. Lastly, none of the available tools offer an intelligent auto-fix feature that automatically suggests the nearest passing color alternative when a chosen pair fails. **Solution:** The application "Color Contrast & Accessibility Checker" is a fully offline, standalone Java Swing application. The application requires the user to input or select two colors, namely, the foreground color (text color) and the background color, and the application will immediately calculate the contrast ratio using the WCAG 2.1 guidelines, show the user an instant preview of the text on the given background, and give the user the result as to whether the given color combination is "Pass" or "Fail" for all three levels of compliance, namely, AA Normal, AA Large Text, and AAA

Impact:

- Ensures all generated palettes meet strict mathematical thresholds for legal and ethical web standards
- Reduces time spent on UI accessibility auditing with auto-generated color suggestions
- Raising awareness of accessibility and inclusivity as a first-class software requirement

Resources:

- IntelliJ(Java IDE)

Conclusion: In conclusion, Color Contrast and Accessibility checker is a real-world tool that addresses a genuine gap in the offline design toolkit. It is a solution to an established problem with measurable criteria (WCAG contrast compliance), has well-defined real-world users (designers and developers), and does so with zero external dependencies.